2019年06月长篇阅读解析第3套

Section B

个人卫星技术的未来——我们准备好了吗?

一、总体分析

本文选自 LiveScience《美国生命科学网》2016. 5. 27 一篇题为 The Future of Personal Satellite Technology is Here(个人卫星技术的未来)的文章。本文指出由于发射成本大幅降低,以立方星(CubeSats)为代表的迷你卫星愈加普及,这在促进太空科学研究的同时也带来了风险,而解决风险的关键思路是培养人们的社会责任感。

二、试题分析

试题	定位词	核心义
36. Given the easier **accessibility**[①] to space, it is time to think about how to prevent misuse of satellites.	[1] the easier accessibility to space [2] it is time to think about	由太空发射日益普及而提出的建议/提醒:应思考如何防止卫星滥用。

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试题	定位词	核心义
37. A group of mini-satellites can work together to **accomplish**② more complex tasks.	[1] A group of mini-satellites [2] more complex tasks	迷你卫星协同合作的好处：完成更加复杂的任务。
38. The greater accessibility of mini-satellites increases the risks of their irresponsible use.	[1] The greater accessibility of mini-satellites [2] the risks of their irresponsible use	迷你卫星越来越容易获得带来的风险：不负责任的使用。
39. Even school pupils can have their CubeSats put in **orbit**③ owing to the lowered **launching**④ cost.	[1] school pupils [2] the lowered launching cost	发射卫星的成本降低带来的影响：甚至小学生都能发射专属卫星。
40. AMSAT is careful about sharing information with outsiders to prevent **hijacking**⑤ of their satellites.	[1] AMSAT [2] sharing information with outsiders [3] hijacking of their satellites	AMSAT 慎与外人分享信息的原因：防止卫星被劫持。
41. NASA offers to launch CubeSats free of charge for educational and research purposes.	[1] NASA [2] to launch CubeSats free of charge [3] for educational and research purposes	NASA 提供免费发射卫星服务的领域：教育和研究。
42. Even with **constraints**⑥, it is possible for some creative developers to take the CubeSat technology in directions that result in harmful outcomes.	[1] constraints [2] some creative developers [3] directions that result in harmful outcomes	某些创新研发者的行为：将 CubeSat 技术用于不良目的。
43. While making significant contributions to space science, CubeSats may **pose hazards to**⑦ other space vehicles.	[1] making significant contributions to space science [2] pose hazards to other space vehicles	CubeSats 利弊兼有：一方面为太空科学做出巨大贡献，另一方面又对其他太空设备造成威胁。
44. Mini-satellites enable operators to study Earth from LEO and space around it.	[1] operators [2] to study Earth from LEO and space around it	迷你卫星的作用：使操作者能够从外太空研究地球及其上空。
45. AMSAT operates on the principle of having all its technical data accessible to its members, preventing the abuse of **amateur satellites**⑧.	[1] AMSAT [2] the principle of having all its technical data accessible to its members	AMSAT 的运行原则：所有技术数据内部分享，其目的是防止业余卫星滥用。

·词汇注释与难句分析·

①**accessibility** [əkˌsesəˈbɪlətɪ] *n.* 可获得性
②**accomplish** [əˈkʌmplɪʃ] *v.* 完成
③**orbit** [ˈɔːbɪt] *n.* (天体等运行的)轨道
④**launch** [lɔːntʃ] *v.* 把(航天器、武器等)发射上天
⑤**hijack** [ˈhaɪdʒæk] *v.* 劫持
⑥**constraint** [kənˈstreɪnt] *n.* 限制，制约
⑦**pose hazards to** 给……带来危害
⑧**amateur satellites** 业余卫星

定位词选取原则：(1)排除多次复现、与主题密切相关的信息表达(多次复现的表达区辨性不足，如 misuse of satellites、their irresponsible use、the abuse of amateur satellites 等；主题相关词因渗透全文且意义相近而难以定位，如(mini-)satellites、CubeSats、CubeSat technology 等，但需注意，如果复现词联合体具有区辨作用，则可选入，如 37 题 A group of mini-satellites、38 题 the risks of their irresponsible use 等)；(2)优先选取具有区辨性的人、物主体(试题中出现次数较少，易于区分，如 school pupils、NASA、some creative developers 等，有些主体虽在试题中复现，但区辨性高可初步锁定试题对应段落，也可重复入选定位词，如 40 和 45 题的 AMSAT)；(3)补充选择其他具有区辨性的细节信息，如 the easier accessibility to space、the lowered launching cost、hijacking of their satellites 等。

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解题思维原则：(略，详见2019年6月第1套)

三、速读巧解

The Future of Personal Satellite Technology is Here
—Are We Ready for it?

[A] ❶Satellites used to be the **exclusive**① **playthings**② of rich governments and wealthy corporations. ❷But increasingly, as space becomes more **democratized**③, they are **coming within reach of**④ ordinary people. ❸Just like *drones* (无人机) before them, **miniature**⑤ satellites are beginning to fundamentally transform our conceptions of who gets to do what up above our heads.

[B] ❶As a recent report from the National Academy of Sciences highlights, these satellites hold tremendous potential for making satellite-based science more accessible than ever before. [38] ❷However, as the cost of getting your own satellite in orbit drops sharply, the risks of irresponsible use grow. ❸The question here is no longer "Can we?" but "Should we?" ❹What are the potential **downsides**⑥ of having a slice of space densely populated by equipment built by people not traditionally **labeled as**⑦ "professionals"? ❺And what would the responsible and beneficial development and use of this technology actually look like? ❻Some of the answers may come from a nonprofit organization that has been building and launching amateur satellites for nearly 50 years.

个人卫星技术的未来
——我们准备好了吗？

人造卫星曾经是富裕国家和有钱企业的专属玩物。但随着太空愈加大众化，人造卫星变得越来越为平常百姓所及。就像之前的无人机一样，微型人造卫星正在彻底改变我们对"谁有机会参与我们头顶之事"的认知。

正如美国国家科学院最近一份报告所强调的那样，在推动卫星科学更加容易获取方面，这些人造卫星拥有巨大潜力。然而，随着往轨道上发射专属人造卫星的成本急剧下降，不负责任地使用卫星的风险也在增加。我们面临的问题已经不是"能做与否"而是"应做与否"。如果某个空间大量充斥着传统上未被称为"专业人士"的人们所制造的卫星设备，会有什么潜在后果呢？对卫星技术负责任且有利的开发和使用到底是什么样的？一个制造和发射业余人造卫星近五十年之久的非营利组织或许能够提供一些解答。

·词汇注释与难句分析·

①**exclusive** [ɪkˈskluːsɪv] *a.* 专有的，独享的
②**plaything** [ˈpleɪˌθɪŋ] *n.* 玩物，玩乐的对象
③**democratized** [dɪˈmɒkrətaɪzd] *a.* 民主化的，大众化的
④**come/be within the reach of** 为……所及
⑤**miniature** [mɪnɪtʃə] *a.* 微型的，小型的
⑥**downside** [ˈdaʊnsaɪd] *n.* 缺点，不利方面
⑦**label sb/sth as** (尤指不公正地)把……称之为，称作

·速读巧解·

[A][B]两段引出本文讨论话题：如今向地球轨道发射个人卫星成本越来越低，由此带来卫星滥用风险。

[A]段介绍如今情况：微型卫星越来越普及，卫星不再是有钱国家和公司的专属。①句与②③句昔今对比：曾经(used to)卫星是有钱国家和公司的专属玩物 VS 如今(becomes/are coming)迷你卫星普及，并改变人们对卫星的认识。**全段核心信息为：**微型卫星的普及。

【定位词定位】无。

[B]段指出问题：微型卫星的容易获取性增加了卫星滥用的风险。①承接[A]段指出微型卫星容易获取有利于普及卫星相关科学知识，②句随即转折指出风险：卫星可能会被滥用。③至⑤句进一步指出卫星滥用将涉及的几个问题：人们是否应该发射个人专属卫星？个人卫星数量激增的风险是什么？何谓负责且有益地使用卫星技术？⑥句初步回答问题：一个非营利组织可能会提供解答。**全段核心信息为：**(1)美国国家科学院报告内容：人造卫星有助于普及卫星科学；(2)微型卫星的风险：滥用。

【定位词定位】38题 The greater accessibility of mini-satellites 概括②句 the cost of getting your own satellite in orbit drops sharply, the risks of their irresistible use 在②句中复现。

【核心义匹配】②句大意"专属卫星成本下降增加了卫星滥用的风险"与38题核心义匹配。

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[C] ❶ Having your personal satellite launched into orbit might sound like an idea **straight out of**① science fiction. ❷ But over the past few decades a unique class of satellites has been created that **fits the bill**②: CubeSats. ❸ The "Cube" here simply refers to the satellite's shape. ❹ The most common CubeSat is a 10 cm cube, so small that a single CubeSat could easily **be mistaken for**③ a **paperweight**④ on your desk. ❺ These mini-satellites can fit in a launch vehicle's formerly "wasted space." [37] ❻ Multiples can be **deployed**⑤ in combination for more complex missions than could be achieved by one CubeSat alone.

[D] [44] ❶ Within their **compact**⑥ bodies these **minute**⑦ satellites are able to house **sensors**⑧ and communications receivers/**transmitters**⑨ that enable operators to study Earth from space, as well as space around Earth. ❷ They're primarily designed for Low Earth Orbit (LEO)—an easily accessible region of space from around 200 to 800 miles above Earth, where human-tended missions like the Hubble Space Telescope and the International Space Station (ISS) **hang out**⑩. ❸ But they can attain more distant orbits; NASA plans for most of its future Earth-escaping **payloads**⑪ (to the moon and Mars especially) to carry CubeSats.

[E] ❶ Because they're so small and light, it costs much less to get a CubeSat into Earth's orbit than a traditional communications or GPS satellite. ❷ For instance, a research group here at Arizona State University recently claimed their developmental small CubeSats could cost as little as $3,000 to put in orbit. [39] ❸ This decrease in cost allows researchers, **hobbyists**⑫ and even elementary school groups to put simple instruments into LEO or even having them deployed from the ISS.

[F] ❶ The first CubeSat was created in the early 2000s, as a way of enabling Stanford graduate students to design, build, test and operate a spacecraft with similar capabilities to *the USSR's Sputnik* (前苏联的人造卫星). ❷ Since then, NASA, the National **Reconnaissance**⑬ Office and even Boeing have all launched and operated CubeSats. ❸ There are more than 130 currently in operation. [41] ❹ The NASA Educational Launch of Nano Satellite program, which offers free launches for educational groups and science missions, is now open to U. S. nonprofit corporations as well. ❺ Clearly, satellites are not just for rocket scientists anymore.

将个人专属卫星发射到卫星轨道上听起来像是直接从科幻小说里搬出来的桥段。但在过去几十年里，人们制造了一类特殊人造卫星就能实现上述要求：立方星。此处，“立方”一词仅仅指的是卫星的形状(是立方体的)。最常见的立方星是一个10厘米见方的立方体，体积非常之小，以至于你很容易会把它误认为是书桌上的镇尺。这些迷你卫星能有效利用运载火箭之前的“浪费空间”。多个迷你卫星联合部署能够执行比单个卫星所能完成的更加复杂的任务。

一些迷你卫星袖珍的身体内能够容纳传感器和通讯接收器/发射器，这能使操作者从太空研究地球以及地球周围的空间。这些卫星被设计主要用于近地轨道——地表上空大约200到800英里的那一部分容易到达的空间区域——此处也是哈勃天文望远镜和国际空间站这些由人类操控的太空任务的所在地。不过，这些卫星也能到达更远的轨道，美国国家航空航天局(NASA)计划在未来逃离地球(特别是飞向月球和火星)的大部分太空装备上携带立方星。

由于它们体型小重量轻，将一颗立方星送上地球轨道的成本要比把一颗传统通讯或GPS卫星送上轨道要便宜得多。例如，亚利桑那州立大学的一个研究团队最近声称，他们所研发的小型立方星只需花费3000美元就能进入轨道。成本下降使得研究人员、业余爱好者、甚至是小学生也能将简单的设备送入近地轨道，甚至能从国际空间站部署这些设备。

第一颗立方星建造于21世纪早期，以供斯坦福大学的研究生设计、建造、测试和操作一台与前苏联人造卫星Sputnik具有相似功能的航天器。此后，美国国家航空航天局(NASA)、国家侦查局甚至波音公司都开始发射并运行立方星。目前，在轨运行的立方星有130多颗。为教育团体和科学任务提供免费发射的NASA纳米卫星发射教育计划，现在也向美国非营利组织开放。显然，人造卫星已经不再是火箭科学家们的专属了。

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[G] [43] ❶ The National Academy of Sciences report emphasizes CubeSats' importance in scientific discovery and the training of future space scientists and engineers. ❷ Yet it also **acknowledges**[14] that widespread deployment of LEO CubeSats isn't risk-free. ❸ The greatest concern the authors **raise**[15] is space **debris**[16]—pieces of "junk" that orbit the earth, with the potential to cause serious damage if they **collide with**[17] operational units, including the ISS.

[H] ❶ Currently, there aren't many CubeSats and they're tracked closely. ❷ Yet as LEO opens up to more amateur satellites, they may pose an increasing threat. ❸ As the report authors point out, even near-misses might lead to the "creation of a **burdensome**[18] **regulatory**[19] framework and affect the future **disposition**[20] of science CubeSats."

国家科学院的报告强调了立方星在科学发现以及培养未来太空科学家和工程师方面的重要性。但它也承认，近地轨道立方星的大量部署并非没有风险。报告作者提出的最大担忧是太空残骸——围绕地球运行的"垃圾"碎片，如果它们与运行设备（包括国际空间站）相撞很可能造成严重损害。

目前，立方星还不是很多，而且它们被密切追踪。但是随着近地轨道向更多业余人造卫星开放，它们可能会带来越来越大的威胁。正如报告作者所指出的那样，甚至近距离的错过也可能导致"一个繁重的监管体系的诞生，并且会影响科学用途的立方星的未来部署"。

·词汇注释与难句分析·

①**straight out of sth** 直接从……出来
②**fit the bill** 刚好符合要求，满足需要
③**be mistaken for sth** 被误认为是某物
④**paperweight** ['peɪpəweɪt] *n.* 镇尺、镇纸
⑤**deploy** [dɪ'plɔɪ] *v.* 部署
⑥**compact** [kəm'pækt] *a.* 微型的，袖珍的
⑦**minute** [maɪ'nju:t] *a.* 微小的
⑧**sensor** ['sensə] *n.* （探测光、热、压力等的）传感器
⑨**transmitter** [trænz'mɪtə] *n.* 发射器
⑩**hang out** 外出闲逛，逗留
⑪**payload** ['peɪləʊd] *n.* （航天器、卫星的）装备
⑫**hobbyist** ['hɒbɪɪst] *n.* 业余爱好者
⑬**reconnaissance** [rɪ'kɒnɪsəns] *n.* （军事）侦查，搜索
⑭**acknowledge** [ək'nɒlɪdʒ] *v.* 承认
⑮**raise** [reɪz] *v.* 提及，提起
⑯**debris** ['deɪbri:] *n.* 碎片，残骸
⑰**collide with** 碰撞，冲突
⑱**burdensome** ['bɜ:dənsəm] *a.* 繁重的，繁冗的
⑲**regulatory** ['regjələtərɪ] *a.* 监管的
⑳**disposition** [dɪspə'zɪʃən] *n.* 排列，布置

Within their compact bodies these minute satellites are able to house sensors and communications receivers/transmitters that enable operators to study the Earth from space, as well as space around the Earth.

Within their compact bodies	these minute satellites	are able to house	sensors and communications receivers/transmitters
地点状语	主语	谓语	宾语

定语从句：that enable operators to study the Earth from space, as well as space around the Earth.

·速读巧解·

[C]到[H]段介绍典型迷你卫星 CubeSat（立方星）的基本信息、易获得性以及带来的风险。

[C][D]段介绍立方星的基本信息。

[C]段引入典型迷你卫星"立方星"，介绍其形状、体积。①句先行铺垫指出发射个人专属卫星曾经只是科幻小说的情节；②句随即转折引出能够把科幻变为现实的"立方星"。③至⑤句介绍立方星的形状、体积等相关信息。⑥句补充指出多个立方星可协同作业完成复杂任务。**全段核心信息为：**立方星的基本介绍。

【定位词定位】37 题 A group of mini-satellites/more complex tasks 同义改写⑤句中 Multiples... in combination/more complex missions.

【核心义匹配】⑥句大意"多个迷你卫星协同合作可完成复杂任务"与 37 题核心义匹配。

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[D]段介绍立方星的用途、运行轨道。①句介绍用途：辅助操作者研究地球及其周围空间。②③句介绍运行轨道：主要在近地轨道(LEO)运行，但也可能达到更远轨道。**全段核心信息为：**(1)立方星的工作原理；(2)立方星的工作范围；(3)NASA 的计划：逃离地球计划大量应用立方星。

【定位词定位】44 题 operators、to study Earth from LEO and space around it 在①句复现。

【核心义匹配】①句大意"迷你卫星能辅助操作者研究地球及其周围空间"与 44 题核心义匹配。

[E]段指出立方星的容易获得性。①句指出由于体积小重量轻的优势，将立方星送入太空成本很低。②句举例说明其发射成本最低仅 3000 美元。③句进而指出发射成本低使得更多人都能发射专属卫星。**全段核心信息为：**(1)立方星发射成本低；(2)亚利桑那州立大学研究成果：立方星发射成本低至 3000 美元；(3)设备发射的普及：小学生也能发射。

【定位词定位】39 题 school pupils 与③句 elementary school groups 同指，the lowered launching cost 同义改写①costs much less、②句 cost as little as $3000、③句 this decrease in cost。

【核心义匹配】③句大意"立方星发射成本下降使小学生都能发射专属卫星"与 39 题核心义匹配。

[F]段介绍立方星的发展起源、目前规模、使用范围等信息。①②句介绍发展起源：出现于 21 世纪初，最早用于辅助斯坦福研究生研究之用；之后 NASA 等国家机构开始参与发射并操控立方星。③句指出现状：超过 130 颗卫星在轨道运行。④句介绍 NASA 纳米卫星发射教育计划用途：不仅用于教育和科研目的，也向非盈利性组织开放。⑤句总结：卫星已经不再是科学家们的专属了。**全段核心信息为：**(1)立方星的最初起源；(2)立方星的应用现状；(3)NASA 纳米卫星发射教育计划的免费发射服务。

【定位词定位】41 题 NASA 在本段复现，to launch CubeSats free of charge、for educational and research purposes 同义改写④句 offers free launches、for educational groups and science missions.

【核心义匹配】④句大意"NASA 为教育和科研团体提供免费发射立方星的服务"与 41 题核心义匹配。

[G][H]段转折指出立方星带来的风险。

[G]段指出立方星的普及使太空垃圾增多。①②句先扬后抑，承上转折指出立方星虽然对科学研究至关重要，但并非没有风险(Yet... CubeSats isn't risk free)。③句具体介绍风险：太空残骸增多，如果撞上太空中的运行设备将造成严重后果。**全段核心信息为：**美国国家科学院报告强调立方星的重要性和风险"撞击太空研究设备"。

【定位词定位】43 题 making significant contributions to space science 概括①句 CubeSats' importance in scientific discovery and the training...；pose hazards to other space vehicles 同义改写③句 cause serious damage if they collide with operational units。

【核心义匹配】①句让步肯定立方星的重要意义，②③句转折指出其带来的风险：增加的太空残骸会撞损其他太空设备，本段主旨与 43 题的核心义匹配。

[H]段具体解释立方星带来的影响。①②句让步转折(Currently, there aren't many Cubesats... Yet...)，指出虽然目前立方星不多，但以后会越来越多，其带来的风险也更大。③句具体说明其带来的影响：给监管规则的制定以及科学用途的立方卫星的未来部署工作带来难题。**全段核心信息为：**立方星的潜在风险日益变大。

【定位词定位】无。

[I] [36] ❶CubeSat researchers suggest that now's the time to **ponder**[①] unexpected and **unintended**[②] possible consequences of more people than ever **having access to**[③] their own **small slice of**[④] space. ❷In an era when you can simply buy a CubeSat kit off the shelf, how can we trust the satellites over our heads were developed with good intentions by people who knew what they were doing? ❸Some "**expert**[⑤] **amateurs**[⑥]" in the satellite game could provide some inspiration for how to **proceed**[⑦] responsibly.

立方星的研究人员建议，是时候应该思考一下越来越多的人能向太空发射专属卫星所带来的可能的意外后果了。在一个轻易就能从货架上买到现成立方星套装的时代，我们如何才能相信我们头顶上卫星的研发者心怀善意且明确知晓自己的所作所为？在这场卫星游戏中，就如何负责任地开展工作这一问题，一些"经验丰富的爱好者"能够给出些许启发。

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[J] ❶In 1969, the Radio Amateur Satellite Corporation (AMSAT) was created in order to foster *ham radio enthusiasts'* (业余无线电爱好者) participation in space research and communication. ❷It continued the efforts, begun in 1961, by Project OSCAR—a U. S. -based group that built and launched the very first nongovernmental satellite just four years after Sputnik. ❸As an organization of volunteers, AMSAT was putting "amateur" satellites in orbit decades before the current CubeSat craze. ❹And over time, its members have **learned a thing or two**[8] about responsibility. ❺Here, open-source development has been a central principle. [45] ❻Within the organization, AMSAT has a **philosophy**[9] of open sourcing everything—making technical data on all aspects of their satellites fully available to everyone in the organization, and when possible, the public. ❼According to a member of the team responsible for FOX 1-A, AMSAT's first CubeSat, this means that there's no way to **sneak**[10] something like **explosives**[11] or an energy emitter into an amateur satellite when everyone has access to the designs and **implementation**[12].

[K] [40] ❶However, they're more cautious about sharing information with nonmembers, as the organization guards against others developing the ability to hijack and take control of their satellites. ❷This form of "self-governance" is possible within long-standing amateur organizations that, over time, are able to build a sense of responsibility to community members, as well as society in general. ❸But what happens when new players emerge, who don't have deep roots within the existing culture?

[L] ❶Hobbyists and students are gaining access to technologies without being part of a long-standing amateur **establishment**[13]. [42] ❷They're still **constrained**[14] by funders, launch providers and a series of regulations—all of which **rein in**[15] what CubeSat developers can and cannot do. ❸But there's a danger they're **ill-equipped to**[16] **think through**[17] potential unintended consequences. ❹What these unintended consequences might be is **admittedly**[18] far from clear. [42] ❺Yet we know innovators can be remarkably creative with taking technologies in unexpected directions. ❻Think of something as seemingly **benign**[19] as the cellphone—we have microfinance and text-based social networking at one end of the **spectrum**[20], and *improvised* (临时制作的) explosive devices at the other.

1969年，无线电爱好者卫星公司(AMSAT)成立了，其目的是鼓励业余无线电爱好者参与太空研究和交流。这家公司延续了1961年开始的OSCAR项目，后者是一家位于美国的组织，它在前苏联人造卫星Sputnik面世四年之后建造和发射了第一颗非政府的卫星。在如今立方星热潮出现的几十年前，AMSAT这家志愿者组织就已经把那些"业余"人造卫星送入了地球轨道。在此期间，它的成员对于责任问题已经了然于心。在这里，开源开发一直都是中心原则。在组织内部，AMSAT有一条开放一切资源的哲学——组织内部的每个人都能完全获取有关卫星方方面面的技术数据，如果可能的话，公众也能获取这些数据。负责AMSAT第一颗立方星FOX 1－A的一位团队成员指出，这意味着在一颗业余卫星上偷藏诸如炸药或者能量发射器的情况不可能出现，因为人人都能获知卫星的设计和操控情况。

然而，在与非组织成员分享信息方面，他们却很谨慎，因为该组织担心其他机构会研发劫持及控制其卫星的技术。这种"自我管理"形式在存在已久的业余爱好者组织内部比较可行，随着时间推移，这些业余组织能够使行业成员以及行业整体建立一种责任感。但是，当那些不甚熟知当前文化的新成员出现后，情况会怎么样呢？

业余爱好者和学生们无需加入一个存在已久的业余组织就能获取这些卫星技术。他们依然会受到投资人、卫星发射供应商和一系列规则的限制——所有这些都约束着立方星研发人员能做什么、不能做什么。但是，他们在充分思考潜在意外后果方面能力欠缺，这很危险。诚然，这些意外后果可能是什么尚不清楚。但是我们很清楚，在将技术应用于意想不到的方向方面，创新者们创意无穷。想想那些看起来和手机一样亲切的东西——一端是小额信贷和基于文本的社交网络，另一端却是临时制作的爆炸装置。

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[M] ❶ This is where a culture of social responsibility around CubeSats becomes important—not simply to ensure that **physical risks**㉑ are minimized, but to engage with a much larger community in anticipating and managing less obvious consequences of the technology. ❷ This is not an easy task. ❸ Yet the evidence from AMSAT and other areas of technology development suggests that responsible amateur communities can and do emerge around novel technologies. ❹ The challenge here, of course, is ensuring that what an amateur community considers to be responsible, actually is. ❺ Here's where there needs to be a much wider public conversation that extends beyond government agencies and **scientific communities**㉒ to include students, hobbyists, and anyone who may potentially stand to be affected by the use of CubeSat technology. [1264 words]

这种情形之下，围绕立方星的社会责任感文化就变得非常重要——不仅要确保将有形风险降至最低，而且在预测和处理不那么明显的技术后果方面还要和更大社群密切合作。这并非易事。但是，AMSAT 以及其他技术开发领域的证据表明，围绕于新技术的负责任的业余社群可能、而且也确实已经出现。当然，目前所面临的难题是确保某个业余社群所认为的负责是真正的负责。这一问题需要更加广泛的社会各界共同讨论，不仅仅涉及政府机构和科研群体，还应该包括学生、业余爱好者以及可能受到立方星技术的使用所影响的任何人。

·词汇注释与难句分析·

①**ponder** ['pɒndə] *v.* 考虑，琢磨
②**unintended** [ʌnɪn'tendɪd] *a.* 无意的，非故意的
③**have access to** 有使用的机会或权利
④**small slice of** 小部分的
⑤**expert** ['ekspɜːt] *a.* 经验丰富的；内行的
⑥**amateur** ['æmətə] *n./a.* 业余爱好者；业余的
⑦**proceed** [prə'siːd] *v.* 继续进行
⑧**learn a thing or two about** 了解/学习很多
⑨**philosophy** [fɪ'lɒsəfɪ] *n.* 生活[工作]准则，指导思想
⑩**sneak** [sniːk] *v.* 偷偷做；偷带
⑪**explosive** [ɪk'spləʊsɪv] *n.* 炸药
⑫**implementation** [ɪmplɪmen'teɪʃən] *n.* 实施，执行
⑬**establishment** [ɪ'stæblɪʃmənt] *n.* 大型组织或机构，企业
⑭**constrain** [kən'streɪn] *v.* 限制
⑮**rein in** 制约，控制
⑯**be ill-equipped to do sth** 做某事能力欠缺的
⑰**think through** 充分考虑
⑱**admittedly** [əd'mɪtɪdlɪ] *adv.* 不可否认，诚然
⑲**benign** [bɪ'naɪn] *a.* 善良的，温顺的
⑳**spectrum** ['spektrəm] *n.* 光谱；范围，幅度
㉑**physical risk** 有形风险
㉒**scientific community** 科学界，科学团体

Here's where there needs to be a much wider public conversation that extends beyond government agencies and scientific communities to include students, hobbyists, and anyone who may potentially stand to be affected by the use of CubeSat technology.

句子主干：Here's（系表结构倒装）where（主语从句引导词）there needs to be a much wider public conversation（there be 句型）

定语从句修饰conversation：that extends beyond government agencies and scientific communities to include students, hobbyists, and anyone

定语从句修饰anyone：who may potentially stand to be affected by the use of CubeSat technology.

·速读巧解·

[I]至[M]段提出解决方案。

[I]段总起指出解决方案的思路：可以参考一些业余专家的做法。①②句发出提醒：如今立方星非常容易获取，我们很难保证卫星不被坏人利用，是时候思考立方星的后果了。③句提出解决思路：一些经验丰富的业余爱好者的做法可能带来启示。**全段核心信息为：**立方星研究者建议：亟待思考卫星发射过多的可能后果。

【定位词定位】36 题 the easier accessibility to space 概括①句 more people than ever having access to...

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space, it is time to think about... 同义改写①句 now's the time to ponder...。

【核心义匹配】①句指出专家建议现在应该考虑立方星所带来的不良后果，②句给出原因"因为立方星方便易得，很难保证不被滥用"，这两句的大意符合36题的核心义。

[J][K]两段具体介绍业余专家机构AMSAT具有责任感的做法以供参考。

[J]段介绍AMSAT的内部开源分享原则。①至③句介绍AMSAT的起源：是由业余无线电爱好者于1969年自愿成立的组织，成立目的是为了促进太空研究和交流，主要业务是发射"业余"卫星。④至⑦句介绍AMSAT的负责任做法(its members have learned... about responsibility)及其好处：所有技术数据全部内部开源共享，成员均能获取卫星设计和操控数据，这能避免卫星为不法分子所利用。**全段核心信息为：**AMSAT的基本信息、理念、以及在责任问题上的丰富经验。

【定位词定位】45题AMSAT在本段复现，the principle of having all its technical data accessible to its members 同义改写⑥句 a philosophy of... making technical data on all aspects of their satellites fully available to everyone in the organization。

【核心义匹配】⑥⑦句的大意"AMSAT内部数据全部共享，避免别有用心者干坏事"与45题核心义匹配。

[K]段指出AMSAT做法的利弊。①句先承上段转折指出AMSAT成员内部共享全部资源，但不轻易对外人分享资源，以防卫星被劫持。②句指出其做法好处：有助于在内部建立责任感；③句指出弊端：新成员没有受到这种责任文化的深度影响，可能会出问题。**全段核心信息为：**AMSAT慎与非组织成员共享信息。

【定位词定位】40题 sharing information with outsiders 同义替换①句 sharing information with nonmembers，hijacking of their satellites 同义替换①句 to hijack... their satellites，AMSAT虽未直接复现，但为本段论述对象。

【核心义匹配】①句大意"AMSAT谨慎与外人分享信息以免卫星被劫持"与40题核心义匹配。

[L]段明确非组织成员可能带来的问题。本段前五句通过连续多次转折(... still... But... admittedly... Yet...)表明非组织成员可能带来的问题：虽无需加入相关成熟组织就能获取技术，但行为仍然受到限制(①②句)→但是由于他们对潜在风险思考不足，很可能会带来危险(③句)→虽然目前风险尚不清楚，但人心难测，很有可能发生意外(④⑤句)。⑥句以手机为例说明⑤句。**全段核心信息为：**(1)非组织成员无法充分考虑发射卫星的潜在后果；(2)技术可能被应用于完全意想不到的不良目的。

【定位词定位】42题 constraints 概括②句 constrained by funders, launch providers...，some creative developers 对应⑤句 innovators... remarkably creative，directions that result in harmful outcomes 概括③句 danger/potential unintended consequences、⑤句 unexpected directions。

【核心义匹配】②⑤句的大意"一些无线电爱好者虽然受到限制，但仍有可能将卫星技术用于不良目的"与42题核心义匹配。

[M]段总结，明确指出培养社会责任意识对解决立方星带来的风险至关重要。①句指出社会责任意识的重要性：不仅将有形风险降至最低，还能防范无形风险。②③句让步转折(This is not an easy task. Yet...)指出：虽然培养社会责任意识困难重重，但却切实可行。④⑤句随即指出培养社会责任意识的关键所在：需要全社会参与讨论。**全段核心信息为：**(1)浪费严重的群体：消费者；(2)浪费的严重程度：全球性问题，美国尤甚。

【定位词定位】无。

四、答案精析

36. 鉴于向太空发射人造卫星变得更加容易，是时候思考如何避免卫星滥用的问题了。

[答案][I] **[精解]**[I]段①句发出提醒：如今立方星非常容易获取，很难保证卫星不被坏人利用，是时候思考立方星的后果了，也即需要思考如何防止卫星被坏人利用。本题是对该两句的概括，其中 how to prevent misuse of satellites 同义转述②句 how can we trust...。

37. 一组迷你卫星协同合作能够完成更加复杂的任务。

[答案][C] **[精解]**[C]段引出立方星这种迷你卫星(mini-satellites)，⑥句指出多个迷你卫星可协同作业完成复杂任务。本题是对⑥句大意的同义改写。

38. 迷你卫星更加容易获取增加了不负责任使用卫星的风险。

[答案][B] **[精解]**[B]段②句指出，如今发射个人专属卫星的成本大幅降低，意味着卫星越来越容易获取，这增加了不负责任地滥用卫星的风险"。本题是对该两句的概括，其中 greater 对应表示变化的动词 drops，均表明卫星日益普及的趋势。

39. 由于发射成本降低，甚至小学生都能将他们的立方星送入轨道。

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[答案][E] [精解][E]段①②句介绍立方星发射成本大幅降低,③句进而指出发射成本低使得包括小学生在内的更多人都能发射专属卫星。本题是对③句内容的同义改写。

40. AMSAT 在与外人分享信息方面非常谨慎,以防他们的卫星被人劫持。

[答案][K] [精解][K]段①句指出 AMSAT 不轻易对外人分享资源,以防卫星被劫持。本题是对该句的同义改写,其中 is careful about 同义替换 are... cautious about;to prevent... 同义替换 guards against...。

41. NASA 为教育和研究目的免费提供发射立方星的服务。

[答案][F] [精解][F]段④句指出 NASA 为教育和科研团队免费提供卫星发射服务。本题是对该句的同义改写。

42. 即便有限制,一些创新的研发者仍然有可能将立方星用于导致不良结果的方向上。

[答案][L] [精解][L]段②句指出新加入的业余卫星爱好者的确受到一些限制,⑤句转而继续指出一些人在利用技术方面创意无穷,很可能会将卫星用于不良目的。本题是对两句内容的总结概括。

43. 立方星尽管为太空科学做出巨大贡献,但也可能给其他太空装置带来危害。

[答案][G] [精解][G]段①句肯定立方星对科学研究至关重要,②③句指出其并非没有风险"太空残骸增多,如果撞上太空中运行的设备将造成严重后果"。本题是对该段大意的概括总结。

44. 迷你卫星使操作者能够从近地轨道研究地球及其周围空间。

[答案][D] [精解][D]段①句介绍立方星(迷你卫星)的用途"研究地球及其周围空间",②句介绍其运行轨道"要在近地轨道(LEO)运行"。本题是对①句的同义概括,LEO 是对句中 space 的具化。

45. AMSAT 的运营原则是,使所有成员都能获取所有技术数据,以避免业余卫星的滥用。

[答案][J] [精解][J]段⑥句介绍 AMSAT 的原则/理念(a philosophy)"所有技术数据全部内部开源共享,人人都能获取卫星设计和操控数据",⑦句指出其好处"避免卫星为不法分子所利用"。本题是对两句内容的概括总结。

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